

Om Shrivastava

Junior Undergraduate

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Academic Qualifications

Year	Degree/Certificate	Institute	CPI/%
2021-2025	B.Tech-Electrical Engineering	Indian Institute of Technology Kanpur	8.8/10
2021	CBSE(XII)	Shiv Jyoti International School,Kota	89.2%
2019	CBSE(X)	K.John Public School, Nagpur	94.6%

Scholastic Achievements

- Secured **Gold medal** in **Inter-IIT Tech Meet 11.0** for the problem statement by Jaguar and Land Rover(JLR). 2023
- Secured **All India Rank 1282** in JEE Advanced & **All India Rank 2758** in JEE Mains among 1 Million candidates.2021
- Selected among **top 300 participants** for Quant Trading Winter School, organized by **Optiver & IIT Delhi**. 2023
- Attained **Gold Level** in World Quant IQC and was offered the **Brain quantitative research consultant** position. 2023

Work Experience

Industry 4.0 | Prof. Nalinaksh S. Vyas | Summer Intern | IIT Kanpur (May'23 - Present)

Objective	Simulate, implement, and drive the adoption of Industry 4.0 principles and technologies in a factory setting.
Approach	Implemented Multilabel classification using transfer learning for defect detection and achieved 100% accuracy . Devised scheduling algorithm on Python , optimizing factory production with inventory data & client needs. Executed Real-Time Machine Monitoring using data from NI-OPC servers on LabView for 15+ machines .
Impact	Enhanced efficiency by 17% through optimized machine scheduling and monitoring, driving productivity.

Key Projects

MUS Rover: Self learning Musical compositional Rules | Prof. Vipul Arora | IIT Kanpur (May'23 - Present)

Objective	To research the possible way to build a model to extract the best compositional rule from the input audio.
Approach	Conducted extensive literature review to find the possibilities to extract the best features from input audio. Utilized Probability theory, Information theory, and Machine learning to Solve optimization problems. Implemented signal processing code to extract the pitch from a dataset of classical music for creating test data
Results	The model produced the best feature distribution and the probability of all the features it extracted.

Jaguar Land Rover's Robotic Charging Challenge | Inter IIT Tech Meet 11.0 | 🏆 (Jan'23-Feb'23)

Objective	Develop a robotic arm for automatically plugging the socket into the charging port of a JLR Electric Vehicle .
Approach	Trained Yolov5 model for detection of a charging port in obstructed views with the help of a custom dataset. Used Transfer Learning based model for pose estimation and trained YOLOv5 model to detect CCS2 port.
Result	Won Gold medal & achieved best arm design & simulation among 19 IITs within budget worth INR 80,000 .

Computer Vision & Sensors of Automi | Mentor:Prof. Venkatesh | Team Humanoid | 🏆 (Dec'22)

- Implemented **Hough transformation** to detect the center of a revolving archery board & calculate it's angular velocity
- Tested and used **FSR-402** and **MPU-6050** with Arduino, which could be used for terrain detection while walking

FPGA and Blockchain | Electronics Club , IIT Kanpur | 🏆 (May'22-July'22)

- Learned **Verilog** through the programming assignment provided based on **half-adder, full adder, and comparator**
- Researched hashing techniques and hash functions that could be implemented for the **Blockchain system**
- Developed a blockchain-based system comprising four operations **transaction, refer, create, delete** in Verilog

Smartwatch with fall detection | Project Mentor | Electronics Club, IIT Kanpur (May'23-Jul'23)

- Mentored **15 mentees** in the development of an affordable smartwatch prototype featuring a fall-detection capability.
- Assisted them in the successful implementation of **TinyML** on Arduino Nano 33 BLE, achieving an accuracy of over **95%**.
- Provided assistance in the integration of **hardware components**, including the **TFT display** and **heart rate sensor**

Desktopography | Project Mentor | Electronics Club, IIT Kanpur | 🏆 (May'23-Jul'23)

- Led a team of **9 members** to implement a **touch feature** on projection using a depth map generated from Intel RealSense
- Generated a **mapping** between projection and computer screen based on detected projection size and computer resolution
- Guided computer to click at mapped position using **PyAutoGUI library** based on a finger lying below a threshold distance.

Technical Skills

Programming Languages	Libraries	Software
C,C++,L ^A T _E X,python	Open CV,Numpy,Pandas,Keras,Matplotlib	Arduino,Edge Impulse

Relevant Courses

(*:Ongoing)

Fundamentals of Computing	Introduction to Machine Learning*	Introduction to Electronics
Microelectronics I	Control Systems Analysis	Introduction to Electrical Engineering
Linear Algebra	Data structure & Algorithms*	Signal systems & network

Positions of Responsibility

Coordinator, Electronics Club (April'23-Present)

- Leading a **two-tier** team structure comprising of **25 secretaries** to foster culture in the domain of electronics.
- Mentored **3 projects** Based on electronics, BCI, and Computer vision in the summers comprising **35+ mentees**.